# Chapter 4 Notes

Note that online corrections for the book can be found at `https://id144254.securedata.net/melvinfitting/errata/book_errors/RevisedSetBookErrors/RevisedSetCorrections.pdf`

## Section 1

### Notes

Note that when Smullyan says: "Thus, for example, the class $V$ is partially ordered under inclusion," by inclusion he means *set inclusion* (that is, being a subset), not *set membership*.

Here is the proof of part (2) of Proposition 1.3. For suppose that some element $x$ had two predecessors $y$ and $z$. Then $y < x$ and $z < x$. If $y < z$, then $y < z < x$, contrary to the fact that $y$ is the predecessor of $x$. Similarly it cannot be that $z < y$. But, because $\leq$ is a linear ordering, either $y \leq z$ or $z \leq y$. In either case, because neither $y < z$ or $z < y$ hold, $y = z$, thus completing the proof.

Note that, for a linear ordering $\leq$ of $A$, all finite non-empty subclasses of $A$ must have a least element. We prove this by induction on the size of the subclass. The base case is obvious (because, by $O_1$, $x \leq x$ for all $x$ in the field of $\leq$). Now suppose the result holds for some $n$. Consider some subclass $S$ of $A$ of size $n+1$. Then for any element $x \in S$, suppose $x$ is not the least element. Then $S - \{x\}$, by the induction assumption, has a least element $a$. Because $x$ is not the least element, $a \leq x$ (by $O_4$), meaning $a$ is the least element of $S$. Thus, for every element $x \in S$, either $x$ is the least element of $S$, or there is some other element least element $a$ of $S$. In any case, $S$ has a least element.

Here is an explicit proof of Proposition 1.7: We can prove by induction that, for a well-ordering $\leq$, every non-empty finite subclass of the field of $\leq$ has a greatest element. The base case, when the subclass has one element, is obvious (because, by $O_1$, $x \leq x$ for all $x$ in the field of $\leq$). Now suppose that, for some non-empty subclass of size $n$, it has a greatest element. Now consider some non-empty subclass $C$ of size $n+1$. By $O_5$, it has a least element $e$. Then $C - \{e\}$ is of size $n$, and thus by the induction assumption, has a greatest element $a$. But because $e$ is the least element of $C$, $e \leq a$. Hence, for all elements $x$ of $C$, $x \leq a$, meaning $a$ is the greatest element of $C$. This completes the induction.

Now having proven this result, suppose that $x$ was a limit element of a well ordering $\leq$ and $L_<(x)$ had finitely many elements. Then, by the above result, $L_<(x)$ must have a greatest element, and thus, by Proposition 1.6, $x$ would not be a limit element, a contradiction.

### Exercise 1.1

Suppose that $\leq$ is a partial ordering such that every non-empty subclass of the field of $\leq$ has a least element. We need to show that, for every $x$ and $y$ in the field of $\leq$, $x \leq y \vee y \leq x$. If $x$ and $y$ are both in the field of $\leq$, then we know that the class $\{x, y\}$ has a least element. If $x$ is the least element, then $x \leq y$, and if $y$ is the least element, then $y \leq x$. Thus either $x \leq y$ or $y \leq x$, therefore completing the proof.

## Exercise 1.2

I do not think the premise of this problem makes sense. For suppose that a linear ordering $\leq$ of $A$ was such that, for every proper lower section $L$ of $A$, there is a least element $x$ of $A$ not in $L$. In fact, we can then prove that $A$ has no proper lower sections. For suppose it has some proper lower section $L$. Then there would be a least element $x$ of $A$ not in $L$. Thus, for every element $y \in L$, $y \leq x$. But, by the definition of a proper lower section, also $x < y$, which contradicts $O_2$. Hence $A$ can have no proper lower sections. Then obviously $\leq$ cannot be a well-ordering of $A$. For if it was, then $A$ would have a least element $e$. But then $\{e\}$ would be a proper lower section of $A$, contrary to the fact that $A$ has no proper lower sections. Thus $\leq$ cannot be a well-ordering of $A$.

Now, if the problem instead said: for every proper lower section $L$ of $A$, there is a least element $x$ of $A - L$, the situation would be different. Here the problem would not be contradictory, but the problem is not true. For instance, we can give the following counterexample. Consider the relation $x \supseteq y$ for all natural numbers $x$ and $y$. Clearly this is a linear ordering. However, there is no natural number that is greater than all others. Thus $\supseteq$ is not a well ordering of the natural numbers. However, it is the case that, for every proper lower section $L$ of $\omega$, there is a least element of $A - L$ (that is, a natural number $n$ of $A - L$ such that $n \supseteq x$ for all $x \in A - L$). For consider some proper lower section $L$ of $\omega$. Then it is a set of natural numbers, and hence has an element $n$ such that $n \subseteq x$ for all $x \in L$. Then $n - 1$ is the least element of $A - L$, that is, $n - 1 \supseteq x$ for all $x \in A - L$. Thus, in fact, the problem does not hold for this particular linear ordering, meaning it is not true.

My doubts were confirmed by looking at the online corrections to the book: This problems was not correct.

Now consider any linear ordering $\leq$ for some class $A$. If we were given (in addition to the premises of the problem), that $A$ itself had a least element $a$, the desired result would follow. For then consider any non-empty subclass $C$ of $A$. If it was the entirety of $A$, then it would have a least element (by assumption). So suppose $C \subset A$. Then let $S$ be the class of all elements of $A$ that are less than every element in $C$. If $S$ is empty, then $a \in C$, in which case $a$ is the least element of $C$. Now suppose $S$ is not empty. Because $S$ is a proper lower section of $A$, it follows that $A - S$ has a least element $e$. But suppose $e \notin C$. Then $e$ would be an element that is $\leq$ all elements in $C$ but not equal to any of them, meaning $e$ is less than all elements of $C$; thus, by the definition of $S$, $e \in S$, contrary to the fact that $e \in A - S$. Thus $e \in C$, meaning $e$ is the least element of $C$. Thus the proof is complete.

## Exercise 1.3

This problem had similar issues as in Exercise 1.2: assumption (c) of the problem implies that there can be no proper lower sections $L$ of $A$ such that $L$ has no greatest element.

So instead I am assuming that assumption (c) means: For any proper lower section $L$ of $A$, if $L$ has no greatest element, then there is a least element of $A - L$.

Suppose that $\leq$ is a linear ordering of $A$.

(1): On the one hand, suppose that $\leq$ is a well ordering of $A$. (a) obviously holds (by $O_5$). As for (b), consider any element $x \in A$ which is not the greatest element in $A$. If the set of all elements of $A$ that are $\geq x$ was empty, then $x$ would be the greatest element of $A$. Thus the set $S$ of all elements of $A$ that are $\geq x$ is non-empty. Because $\leq$ is a well ordering, $S$ must contain a least element $e$. Then obviously $e$ is the successor of $x$. As for (c), consider any proper lower section $L$ of $A$ such that $L$ has no greatest element. Because it is a proper lower section, $A - L$ is non-empty, and hence, because $\leq$ is a well ordering, contains a least element. This completes the first part of the proof.

(2): On the other hand, suppose that conditions (a-c) all hold. We must show that $\leq$ is a well ordering for $A$. By (a), $A$ has some least element $a$. Now consider any non-empty subclass $C$ of $A$. If $C = A$, then it would have a least element (by assumption). So suppose $C \subset A$. Then let $S$ be the class of all elements of $A$ that are less than every element in $C$. If $S$ is empty, then $a \in A$, in which case $a$ is the least element of $C$. Now suppose $S$ is not empty. Clearly $S$ is a proper lower section of $A$. If $S$ had no greatest element, then, by (c), $A - S$ would contain a least element $e$ (and of course $e \in C$). On the other hand, suppose that $S$ did contain a greatest element $b$ (and $b$ cannot be the greatest element of $A$). Then, by (b), it has a successor $c$. Of course, $c$ cannot be in $S$, as that would contradict the fact that $b$ was the greatest element of $S$. First, $c$ must be in $C$. For suppose it weren't. Then there is some element $x \in C$ such that $x < c$ (because if this weren't true, then $c$ would be in $S$). Then $b < x < c$, contrary to the fact that $c$ is the successor of $b$. Thus $c \in C$. Second, $c$ is the least element of $C$. For suppose it weren't. Then there would be some element $y \in C$ such that $y < c$. But then $b < y < c$, contrary to

the fact that $c$ is the successor of $b$. Thus $c$ is the least element of $C$. Thus every non-empty subclass of $A$ contains a least element, meaning $\leq$ is a well ordering of $A$.

## Exercise 1.4

We prove that in a well ordering, all backward paths must be finite. What this amounts to saying is that, if $\leq$ is a well-ordering, then there can be no denumerable backward paths. So suppose there was some denumerable backwards path $a_0, a_1, ..., a_n, ....$ Because this is a well-ordering, the class $\{a_0, a_1, ..., a_n, ...\}$ has a least element. Thus for some $m$, it must be the case that $a_m$ is the least element of $\{a_0, a_1, ..., a_n, ...\}$. Because the sequence is denumerable, there must be a term $a_{m+1}$ of the sequence. By the definition of a backward path, $a_{m+1} < a_m$. But also $a_m \leq a_{m+1}$, which is a contradiction (by $O_2$, as this would imply $a_m = a_{m+1}$, but because $a_{m+1} < a_m$, $a_m \neq a_{m+1}$).

## Exercise 1.5

(a): On the one hand, we assume the first form of the AC. Now consider any non-empty set $A$ of pairwise disjoint non-empty sets. By the AC, there is some function $f$ such that $f(x) \in x$ for all elements $x \in A$. Then take $C$ to be the class of all $x$ such that $x = f(y)$ for some $y \in A$. Because $A$ is a set of pairwise disjoint sets, and $f(y) \in y$ for every $y \in A$, it follows that $C$ contains exactly one element of each element of $A$.

(b): On the other hand, suppose that the second form of the AC holds. Now consider any non-empty set $A$ of non-empty sets. Note that for any set $x$, $\{x\}$ is a non-empty set of pairwise disjoint non-empty sets, and hence, there exists some set $C$ that consists of exactly one element of $x$. Denote this set $C$ for this set $x$ as $C_x$, and let $a_x$ be the one element of $C_x$. Then take the function $f$ to be the set of all ordered pairs $\langle x, a_x \rangle$, which is clearly a function such that for each $x$ in $A$, $f(x) \in x$.

## Exercise 1.6

It suffices to show that this form is equivalent to the first form, and then, by Exercise 1.5, it follows that it is also equivalent to the form in that problem.

(a): On the one hand, suppose that the first form of the AC holds, and consider some non-empty set $A$ of non-empty sets. Then, by the AC, there is some function $f$ such that $f(x) \in x$ for all elements $x$ of $\mathcal{P}(A) - \{A\}$ (that is, the set of all proper subsets of $A$), that is, for all proper subsets $x$ of $A$, $f(x) \in x$.

(b): Now suppose that the second form (the one presented in this problem) of the AC holds, and consider some non-empty set $A$ of non-empty sets. Then $\cup A$ is a set, and hence, by the AC, there is some function $f$ such that $f(x) \in x$ for all proper subsets of $\cup A$. Then we take the function $g$ to be the set of all ordered pairs $\langle x, f(x) \rangle$ for all elements $x$ of $A$, which works, as $x$ is a subset of $\cup A$.

## Exercise 1.7

We wish to prove that every infinite set can be put into a 1-1 correspondence with a proper subset of itself (assuming the AC). So, consider any infinite set $A$. Then, by exercise 1.8, it has a denumerable subset $D$, and there is a 1-1 function $f$ from the natural numbers to elements of $D$. Pick some element of $D$ ($f(0)$, say). Then $D - f(0)$ is still a denumerable subset. Now we can define a 1-1 function $g$ from $A$ to $A - \{f(0)\}$ (a proper subset of $A$) as follows: Assume $x \in A$. If $x \in D$, then $x = f(n)$ for some $n$, and we add the ordered pair $\langle f(n), f(n^+) \rangle$ to the function $g$. If $x \notin D$, then we add the ordered pair $\langle x, x \rangle$ to $g$. Thus this a 1-1 function $g$ from $A$ onto $A - \{f(0)\}$, therefore completing the proof.

### Exercise 1.8

It can easily be shown by induction that every infinite set $A$ contains an infinite amount of subsets of size $n$ for every $n > 0$. The base case, when $n = 1$, holds, as, because $A$ is infinite, there is, for every element $x$ of $A$, the subset $\{x\}$ of $A$, and there are infinitely many of these. Now suppose the result holds for some $n$, that is, $A$ has an infinite amount of subsets of size $n$. For each of these subsets $S$ of size $n$, $A - S$ (which must also be infinite) has an infinite number of subsets of size 1 (by having proven the base case), and then, for each element $x \in A - S$, the subset $S \cup \{x\}$ is of size $n + 1$, and because there are infinitely many of these (because $A - S$ contains infinitely many elements), the induction is complete.

Now consider any infinite set $A$. Then the class $C$ of all subsets of $A$ of size one is a set (because it is a subset of $\mathcal{P}(A)$). Then $\{C\}$ is a set, meaning, by the AC, there is some function $f_0^0$ such that $f_0^0(C) \in C$, that is, $f_0^0(C)$ is a subset of $A$ of size one. Because $f_0^0(C)$ is a set, so is $\{f_0^0(C)\}$ (which is a non-empty set of non-empty sets), meaning, by the AC, there is some function $f_1^0$ such that $f_1^0(f_0^0(C)) \in f_0^0(C)$, that is, $f_1^0(f_0^0(C))$ is an element of $A$ ($f_1^0(f_0^0(C))$ is an element of a subset of $A$ of size one). Then we can establish a denumerable sequence $a_0, a_1, ..., a_n, ...$ of elements of $A$ as follows: Take $a_0$ to be $f_1^0(f_0^0(C))$, and let $S_0 = A$ and $C_0 = C$. Then let $S_1 = S_0 - \{a_0\}$ and $C_1$ be the set of all subsets of $S_1$ of size one, and, in a similar manner as in how we obtained $f_1^0(f_0^0(C))$, we can define functions $f_1^1$ and $f_0^1$ using the AC such that $f_1^1(f_0^1(C_1))$ is an element of $S_1$ not equal to $a_0$ (because $S_1$ does not contain $a_1$). Then take $a_1$ to be $f_1^1(f_0^1(C_1))$. In general, if we have $a_i$ and $S_i$ for all numbers $i$ less than or equal to $n$, we take $S_{n+1}$ to be $S_n - \{a_n\}$, $C_{n+1}$ to be the set of all subsets of $S_{n+1}$ of size one, and we can define functions $f_1^{n+1}$ and $f_0^{n+1}$ using the AC such that $f_1^{n+1}(f_0^{n+1}(C_{n+1}))$ is an element of $S_{n+1}$ not equal to $a_i$ for all $i \leq n$. Obviously, each $a_i$ is distinct from all others in the sequence. Thus the set of all $a_i$ is a denumerable subset of $A$.

## Section 2

### Notes

Note that $\omega$ is not superinductive (under the successor function). This is because it is not the case that $\omega$ is closed under chain unions. For instance, consider the subset $C : \{1, 2, ..., n, ...\}$ of $\omega$, which is obviously a chain. We show that it is not the case that $\cup C \in \omega$. For suppose $\cup C \in \omega$. Then, by definition of $\omega$, $\cup C$ would be some natural number $n$. Obviously $n \neq 0$. Now, $n^+ \notin n$ (by Theorem 5.6 of Chapter 3). But $n^{++}$, which includes $n^+$, is an element of $C$; thus $n^+$ is an element of $\cup C$, a contradiction. Hence $\cup C \notin \omega$.

Here is an explanation of why $M$ is minimally superinductive under $g$ in Theorem 2.8. We know that $M$ is the intersection of all subsets of $S$ that are superinductive under $g$. Now, let $C$ be the set of all subsets of $S$ that are superinductive under $g$. Then $M = \cap C$. Because every element of $C$ is superinductive under $g$, every element of $C$ is closed under $g$, and every element of $g$ is closed under chain unions. Thus, by Lemma 2.7, $\cap C$, that is, $M$, is closed under $g$ and is closed under chain unions. Obviously $\emptyset \in M$ (because $\emptyset$ is an element of all superinductive sets, and hence of all elements of $C$). Lastly, we must show that $M$ is minimally superinductive under $g$. Suppose that some proper subset $A$ of $M$ was superinductive under $g$. Thus there must be some element $a \in M$ such that $a \notin A$. Now, clearly $A$ must be a subset of $S$. Thus $A$ is a subset of $S$ which is superinductive under $g$. But then $A \in C$, and because $a \in M$ ($a \in \cap C$), $a \in A$, a contradiction. Thus $M$ must be minimally superinductive under $g$ (and $M$ is a set because $M \subseteq S$).

Here is why $\cup M \in S$ in the proof of Theorem 2.8. As remarked in the proof, we know that $\cup M \in M$. But $M$ is the intersection of all subsets of $S$ that are superinductive under $g$. But because $S$ itself is superinductive under $g$ (and is a subset of itself), $\cup M \in M$ implies that $\cup M \in S$.

There are multiple things to be noted for the proof of Theorem 2.10. First, the definition of $S_b$ is not correct. For, given the current definition, we cannot definitively know that $S_b$ is closed under $g'$. For suppose that $x \in S_b$. Then either $x \neq \emptyset$ or $x = \emptyset$. In the first case, we know that $g'(x) = g(x)$, and because $x \subseteq g(x)$ (because $g$ is progressing), $x \subseteq g'(x)$. Thus, because $x$ is a set which includes $b$ (by the definition of $S_b$), $g'(x)$ also includes $b$, meaning $g'(x) \in S_b$. However, if $x = \emptyset$, then $g'(x) = b$. But there is no guarantee that $b \in S_b$: it very well could be the case that $b \notin b$. In fact, it very well could be that $S_b$ (defined in this way) only contains the empty set!

I will remark that this (incorrect) definition of $S_b$ does make $S_b$ closed under chain unions. For consider any chain $C$ of elements of $S_b$. Then consider $\cup C$. Each element of $C$ is either $\emptyset$ or a set $x$ such that $b \in x$. If $C$ only contains $\emptyset$, then $\cup C = \emptyset$, and thus $\cup C \in S_b$. Otherwise, $C$ contains some element $x$ such that $b \in x$. But then $\cup C$ contains all elements of $x$,

and hence contains $b$, meaning $b \in \cup C$, and thus $\cup C \in S_b$.

The proof of Theorem 2.10 which I found is very similar to the one given in the book, just with a slightly different definition of $S_b$:

**Corrected Proof of Theorem 2.10**:
Assume the hypothesis. As in the original proof, the case when $b = \emptyset$ is just Thoerem 2.8; so assume $b \neq \emptyset$. We let $S_b$ be the set of all elements $x$ of $S$ for which $b \subseteq x$ (and this *does* include $b$), together with $\emptyset$. This is closed under chain unions for the following reason: Consider some chain $C$ of elements of $S_b$. Then consider $\cup C$. Each element of $C$ is either $\emptyset$ or a set that $b$ is a subset of. If $C$ only contains $\emptyset$, then $\cup C = \emptyset$, and thus $\cup C \in S_b$. Otherwise, $C$ contains some element $x$ such that $b \subseteq x$. But then $\cup C$ contains all of the elements of $x$, and thus also all of the elements of $b$, meaning $b \subseteq \cup C$, and thus $\cup C \in S_b$. Thus $S_b$ is closed under chain unions. Then we define $g'$ in the same way as in the book: $g'(\emptyset) = b$, and for any $x \in S_b$ other than $\emptyset$, take $g'(x) = g(x)$. Then $g'$ is progressing for the following reason: For consider any $x \in S_b$. If $x = \emptyset$, then obviously $x \subseteq g'(x)$. Otherwise, if $x \neq \emptyset$, then $g'(x) = g(x)$, and because $x \subseteq g(x)$ (because we assumed that $g$ was progressing), $x \subseteq g'(x)$. Thus $g'$ is progressing. Next, we need to show that $S_b$ is closed under $g'$. For suppose that $x \in S_b$. If $x = \emptyset$, then $g'(x) = b$, and because $b \in S_b$, $g'(x) \in b$. Now suppose $x \neq \emptyset$. Because $x \in S_b$ (and $x \neq \emptyset$), $b \subseteq x$. Thus, because $x \subseteq g'(x)$, $b \subseteq g'(x)$, meaning (by the definition of $S_b$), $g'(x) \in S_b$. Thus $S_b$ is closed under $g'$. Lastly, here is why $S_b$ is a set: $S_b - \{\emptyset\}$ is a set because it is a subset of $S$ (which is assumed to be a set), and thus $\cup\{(S_b - \{\emptyset\}), \{\emptyset\}\}$, that is, $S_b$, is also a set. Thus $S_b$ satisfies the criteria of Theorem 2.8, meaning there is some element $x$ of $S_b$ such that $g'(x) = x$. Now, it cannot be that $x = \emptyset$, or else $b = g'(x) = x = \emptyset$. Thus $x \neq \emptyset$, meaning (by the definition of $S_b$) $b \subseteq x$. Also, because $x \neq \emptyset$, $g'(x) = g(x)$, so that $g(x) = x$. Thus $b \subseteq x$ and $g(x) = x$, therefore completing the proof (and obviously $x \in S$).

It very well could be that the book simply contained a typo, and meant to define $S_b$ in the way it was defined in the correct proof.

### Exercise 2.1

Consider any set $S$:

(1) Suppose that every element of $S$ is closed under a function $g$. Now consider $\cap S$. To prove that $\cap S$ is also closed under $g$, we need to show that if $x \in \cap S$, then $g(x) \in \cap S$. So suppose $x \in \cap S$. Then $x$ is an element of every element of $S$. But every element of $S$ is closed under $g$. Thus $g(x)$ is an element of every element of $S$, meaning $g(x) \in \cap S$.

(2) Suppose that every element of $S$ is closed under chain unions. Now consider $\cap S$. To prove that $\cap S$ is also closed under chain unions, we need to show that for any chain $C$ of elements of $\cap S$, $\cup C \in A$. So consider some chain $C$ of elements of $\cap S$. Thus $C$ is also a chain of elements of every element of $S$. But, because every element of $S$ is closed under chain unions, $\cup C$ is an element of every element of $S$. Thus $\cup C \in \cap S$.

## Sections 3 and 4

### Notes

In the proof of Theorem 3.1, there is no justification given for why $L$ is a set. Here is justification for this: By definition, $L$ is the class of all elements $x$ of $M$ that are proper subsets of all elements of $A$ (a subclass of $M$).If $L$ is empty, then obviously it is a set. On the other hand, suppose that $L$ is non-empty. Now, every element of $A$ is a set. Thus each element of $L$ is an element of $\mathcal{P}(x)$ for every $x \in A$. Thus $L$ is just a subset of $\mathcal{P}(x)$ for every $x \in A$, meaning $L$ is a set (as $\mathcal{P}(x)$ is a set for every $x \in A$).

In Theorem 3.1, here is why $M$ is linearly ordered under inclusion. In fact, every class is partially ordered under inclusion. By theorem 2.3, $O_4$ is also satisfied, meaning $M$ is linearly ordered under inclusion.

Note that $\emptyset$ can be well ordered under $\emptyset$.

In the discussion of the proof of Theorem 4.4, here is why $a \leq b$ iff either $a = b$ or $a$ comes into $A$ before $b$. On the one hand, suppose that $a \leq b$. Also suppose $a \neq b$. Then $F(a) \subset F(b)$. If $F(a)$ contained $b$, then $F(a)$ would be a lesser

element of $N$ than $F(b)$ which contains $b$, a contradiction. Thus $F(a)$ must not contain $b$. On the other hand, suppose that either $a = b$ or $a$ comes into $A$ before $b$. If $a = b$, obviously $a \leq b$. Now suppose $a$ comes into $A$ before $b$. Thus $F(a)$ does not contain $b$. Because $\leq$ is a well ordering, either $a \leq b$ or $b \leq a$. If $a \leq b$ did not hold, then we would have $b < a$, in which case $F(b) \subset F(a)$, meaning $F(a)$ did contain $b$, a contradiction. Thus $a \leq b$.

Consider some class $A$ that is well-ordered under $\subseteq$. It then follows that $L_<(x)$ is a set for all $x \in A$. For consider any $x \in A$. Then either $L_<(x)$ is empty or it is not. If it is empty, then $L_<(x) = \emptyset$, which is a set. If $L_<(x)$ is not empty, note that $L_<(x)$ is just the class of all elements of $A$ that are subsets of $x$. Thus all elements of $L_<(x)$ are subsets of $x$, meaning each element of $L_<(x)$ is an element of $\mathcal{P}(x)$ (as $x$ is a set). Thus $L_<(x)$ is just a subset of $\mathcal{P}(x)$, meaning it is a set.

## Exercise 4.1

(a): On the one hand, suppose that $a \leq b$. Thus $F(a) \subseteq F(b)$. Now consider some element $x$ of $N$ that contains $b$. It cannot be that $x \subset F(a)$. For if that were true, then $x$ would be an element of $N$ less than $F(b)$ that contains $b$, a contradiction. Thus $F(a) \subseteq x$, meaning $x$ contains $a$. On the other hand, suppose that every element of $N$ that contains $b$ also contains $a$. Now consider $F(a)$ and $F(b)$. If $F(b) \subset F(a)$, then, because $F(b)$ was assumed to contain $a$, $F(b)$ would be less than $F(a)$, contrary to the fact that $F(a)$ is the least element of $N$ that contains $a$. Thus $F(a) \subseteq F(b)$, meaning $a \leq b$.

(b): This problem has a typo. It should say $x \in \cup N$, not $x \in N$: Suppose $x \in F(a)$ for some $x \in \cup N$ and $a \in \cup N$. Either $a \leq x$ or $x \leq a$. If $a < x$, then $F(a) \subset F(x)$. But it was assumed that $x \in F(a)$; thus $F(a)$, which contains $x$, would be less than $F(x)$, a contradiction. Thus $x \leq a$.

(c): Suppose $a \in \cup N$, and that $a$ is not the greatest element of $\cup N$. Now consider the successor $b$ of $a$ (and it must have some successor, as it is the least element of the class of all elements greater than $a$, which exists as this is a well ordering). For every $x \in \cup N - F(a)$, either $x \leq b$ or $b \leq x$. Suppose that $x < b$. Then $F(x) \subset F(b)$. Now, either $F(x) \subseteq F(a)$ or $F(a) \subseteq F(x)$. If $F(a) \subset F(x)$, this would mean $F(a) \subset F(x) \subset F(b)$ and thus $a < x < b$, contrary to the fact that $b$ is the successor of $a$. Thus $F(x) \subseteq F(a)$, meaning $x \in F(a)$. But $\cup N - F(a)$ contains no elements of $F(a)$. Thus $x$ is not in $\cup N - F(a)$. Hence for no element $x$ in $\cup N - F(a)$ is it the case that $x < b$, meaning $b \leq x$ for all all $x \in \cup N - F(a)$, and thus $b$ is the least element of $\cup N - F(a)$.

(d): This problem has a typo: It should say $a \in \cup N$, not $a \in N$. Suppose that $b$ is a successor element of $N$, and the predecessor of $b$ is $m$. We are to show that $b = F(a)$ for some $a \in \cup N$. Note that, because $N$ is slowly well ordered under $\subseteq$, $b$ contains exactly one more element $e$ than $m$. We can prove that $F(e) = b$. For suppose that there was some $y \in N$ that contained $e$ such that $y \subset b$. Now, either $y \subseteq m$ or $m \subseteq y$. But, because $e \notin m$, $m \subset y$. Thus $m \subset y \subset b$, contrary to the fact that $b$ is the successor of $m$. Thus, for all $y \in N$ such that $y$ contains $e$, $b \subseteq y$, meaning $F(e) = b$.

Here is why part (d) proves that $F$ is a 1-1 function from $\cup N$ to $M$. We need to show that $x$ is in the range of $F$ iff it is in $M$. Part (d) proves that, if it is in $M$, then it is in the range. On the other hand, suppose that $x$ is in the range of $F$, meaning $x = F(a)$ for some $a$ and $a \in F(a)$. It cannot be the least element of $N$, as it was assumed that $\emptyset$ is the least element of $N$, and $a \notin \emptyset$. Also, $x$ cannot be a limit element. For if it were, it was assumed that this would mean that $x$ is the union of the set of its lesser elements. But then $a$ would be a member of a lesser element of $x$, contrary to the assumption that $x = F(a)$. Thus, because $x$ is neither the least element of $N$ nor a limit element of $N$, it must be a successor element. Thus $x$ is in the range of $F$ iff $x \in M$. Then we need to show that, if $F(a) = F(b)$, then $a = b$. But this was shown already in the proof of Theorem 4.4. Thus we have shown that $F$ is a 1-1 function from $\cup N$ onto the class $M$.

It is easy to see that if $a \in \cup N$ and $b \in \cup N$ and $b$ is the successor of $a$, then, for any $x \in N$, if $F(a) \subset x \subset F(b)$, then $x$ is a limit element. For, $x$ obviously cannot be the least element (as it is assumed to be $\emptyset$). Also, if $x$ were a successor element, then it would be $F(c)$ for some $c$. Thus $F(a) \subset F(c) \subset F(b)$, in which case $a < c < b$, contradicting the fact that $b$ is the successor of $a$. Thus $x$ is a limit element.

## Exercise 4.2

Suppose that for some set $S$, there is a choice function $f$ for $\mathcal{P}(S)$. We define the choice function $g$ for $S$ as follows: Consider some subset $C$ of $S$. The set $F$ of all elements $y$ such that $y = \{x\}$ for some $x \in C$ is a subset of $\mathcal{P}(S)$. Thus $f(F) \in F$, that is, $f(F) = \{x\}$ for some $x \in C$. Then we take $g(C)$ to be $x$. Thus $g$ is a choice function for $S$.

### Exercise 4.3

This problem is incorrect (per the online corrections). If we were given the added condition that every element of $S$ had a choice function, then we are able to prove this. For suppose that $S$ has a choice function $f$ and every element $x$ of $S$ has a choice function $f_x$. We can define a choice function $g$ for $\cup S$ as follows: Consider any subset $C$ of $\cup S$. Now consider the set $A$ of all elements $x$ of $S$ for which at least one element of $C$ is an element of $x$. Because $A$ is a subset of $S$, we can choose an element $f(A)$ of $A$. Now consider the set $G$ of all elements of $f(A)$ that are elements of $C$. Because $G$ is a subset of $f(A)$, and $f(A)$ was assumed to have a choice function $f_{f(A)}$, we can choose an element $y = f_{f(A)}(G)$ such that $y \in G$, that is, $y \in C$. Thus we take $g(C)$ to be $y$.

## Section 5

### Notes

Here is why in the proof of Proposition 5.10, $\cup C$ not being a maximal element of $S$ would imply that $C$ would be a proper subset of $C \cup \{x\}$ for any $x$ for which $\cup C \subset x$. This is to the effect of saying that $x \notin C$. Now, suppose $x \in C$. Then $x \subseteq \cup C$, contrary to the assumption that $\cup C \subset x$. Thus $\cup C$ must be a maximal element of $S$.

### Exercise 5.1

Assume Hausdorff's maximal principle. Thus, for every non-empty set $A$, the set $S$ of all chains of elements of $A$ is of type $M$. We are to prove the axiom of choice. Given a non-empty set $S$, let $\Sigma$ be the set of all choice functions of all non-empty subsets of $S$. It is a set because it is a subset of $\mathcal{P}(S) \times S$. Clearly $\Sigma$ is non-empty (as all finite subsets of $S$ must have a choice function). Next, we can show that $\Sigma$ is closed

I could not see how to prove the result with the given definition of $\Sigma$ (in particular, I could not show that it was closed under chain unions). However, I found a proof using a slightly different definition of $\Sigma$. First, define a *semi-choice function* for a set $S$ to be a function $f$ such that $f(x) \in x$ for all $x$ in some subset of $\mathcal{P}(S)$. That is, a semi-choice function is a function from a subset of $\mathcal{P}(S)$ into $S$. Thus a choice function is also a semi-choice function. Now we prove the theorem:

Assume Hausdorff's maximal principle. Now consider any non-empty set $S$. Let $\Sigma$ be the set of all semi-choice functions for all subsets of $S$ (it is non-empty, as, for example, all finite subsets of $S$ have choice functions). $\Sigma$ is a set, as it is a subset of $\mathcal{P}(\mathcal{P}(S) \times S)$. Next we show that $\Sigma$ is closed under chain unions. So consider some chain $C$ of elements of $\Sigma$. Now consider $\cup C$. Each element of $\cup C$ is of the form $\langle x, y\rangle$ where $x$ is a subset of $S$ and $y \in S$. Then consider the set $F$ of all elements $x$ such that $x \in y$ for at least one $y$ such that $\langle y, z\rangle \in \cup C$ for at least one $z$. Obviously, $F \subseteq S$. We show that $\cup C$ is a semi-choice function for $F$, which is to say, if $x$ is in the domain of $\cup C$, then $x$ is a subset of $F$, and if $y$ is in the range of $\cup C$, then $y \in F$ (and that, if $\cup C(x) \neq \cup C(y)$, then $x \neq y$). For suppose $x$ is in the domain of $\cup C$. Then obviously $x$ is a subset of elements of $F$. Now suppose $y$ is in the range of $\cup C$. Then, for some $x \subseteq F$, $\langle x, y\rangle \in \cup C$ such that $y \in x$. Thus $y \in F$. Lastly, suppose $\cup C(x) \neq \cup C(y)$. We need to show that $x \neq y$. For some $n \in C$, $\langle x, \cup C(x)\rangle \in n$, and for some $m \in C$, $\langle y, \cup C(y)\rangle \in m$. Becuase $C$ is a chain, $m$ and $n$ are comparable. Without loss of generality, suppose $m \subseteq n$. Thus $y$ is in the domain of $n$ and $n(y) = m(y) = \cup C(y)$. Also $n(x) = \cup C(x) \neq \cup C(y)$. Then, because $n$ is a function, it follows that $x \neq y$. Thus, we have shown that $\cup C$ is a semi-choice function for some subset of $S$, meaning $\cup C \in \Sigma$. This proves that $\Sigma$ is closed under chain unions.

Having shown the above, recall that Hausdorff's principle is equivalent to Kuratowski's maximal principle (which is that every set closed under chain unions is of type $M$). Thus, because we assumed Hausdorff's principle, and we have shown that $\Sigma$ is closed under chain unions, it follows that $\Sigma$ is of type $M$, that is, $\Sigma$ has some maximal element $f$. We can show that $f$ is a choice function for $S$. First, by definition of $\Sigma$, $f$ is a semi-choice function for some subset $A$ of $S$. We show that, for whatever subset $A$ of $S$ that $f$ is a semi-choice function for, $f$ is in fact a choice function for $A$. For suppose it weren't. Then there would be some non-empty subset $x$ of $A$ for which $f$ was not defined. Then there exists some $a \in x$, and we let $g$ be the semi-choice function of $A$ with all of the same ordered pairs as $f$, as well as the ordered pair $\langle x, a\rangle$. Then $f$ is a subset of $g$, contrary to the fact that $f$ is a maximal element of $\Sigma$. Thus $f$ must be a choice function for $A$. Now, clearly $f$ is a semi-choice function for $S$ (as all semi-choice functions of subsets of $S$ are also semi-choice functions for $S$ itself). Thus, we have shown that $f$ is a choice function for $S$, therefore completing the proof.